

DRAWINGS AND FIGURE DESCRIPTIONS

Overall Architecture Block Diagram of the PNetGimini System

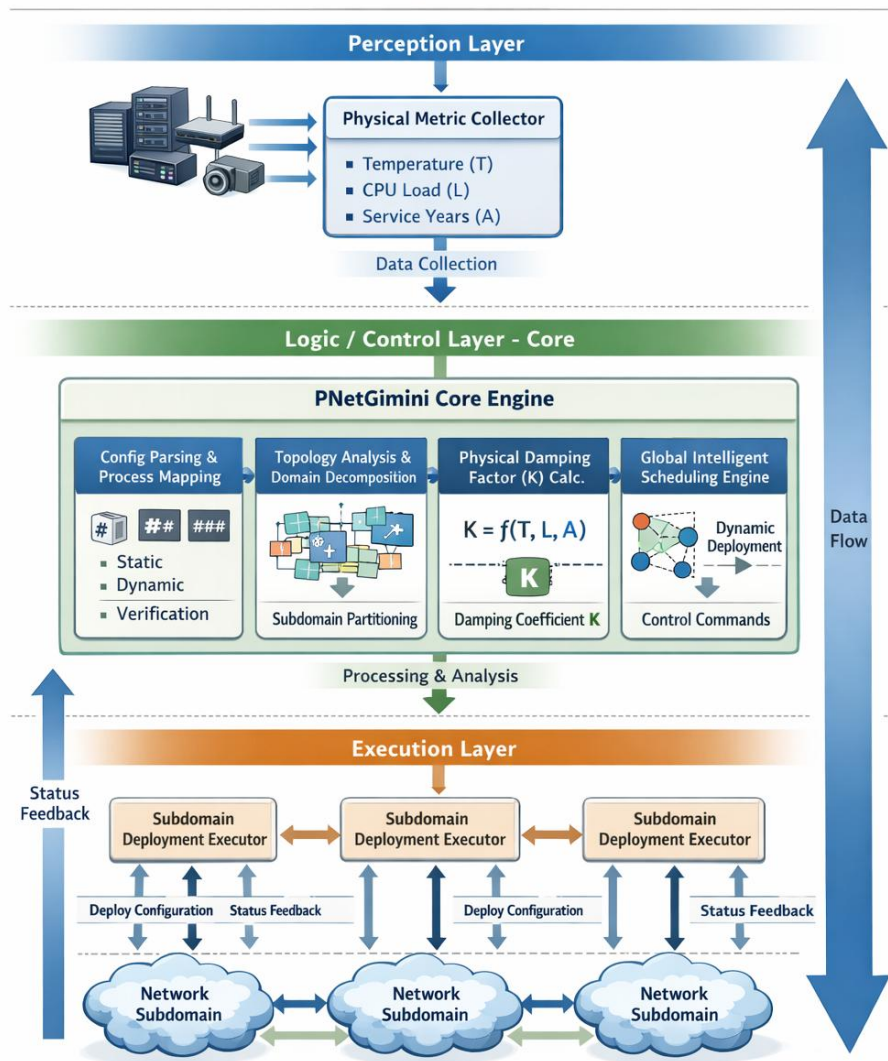


Figure 1: Overall Architecture Block Diagram of the PNetGimini System

As shown in the figure, the system adopts a three-layer architecture design:

1. **Perception Layer (Upper):** Real-time physical state data such as temperature (T), CPU load (L), and service years (A) of network devices are acquired through a Physical Metrics Collector.
2. **Logic Layer (Middle, PNetGimini Core Engine):** Receiving data from the Perception Layer and the raw configuration, it processes them sequentially through modules including Configuration Parsing and Process Step Mapping, Topology Analysis and Domain Decomposition, and Physical Damping Factor Calculation. Finally, a Global Scheduler generates the dynamic deployment strategy.

3. **Execution Layer (Lower):** Multiple Subdomain Deployment Executors work in parallel. Based on scheduling instructions, they perform configuration deployment and state management on their respectively assigned network subdomains, and feed back the execution status to the Perception Layer, forming a closed-loop control.

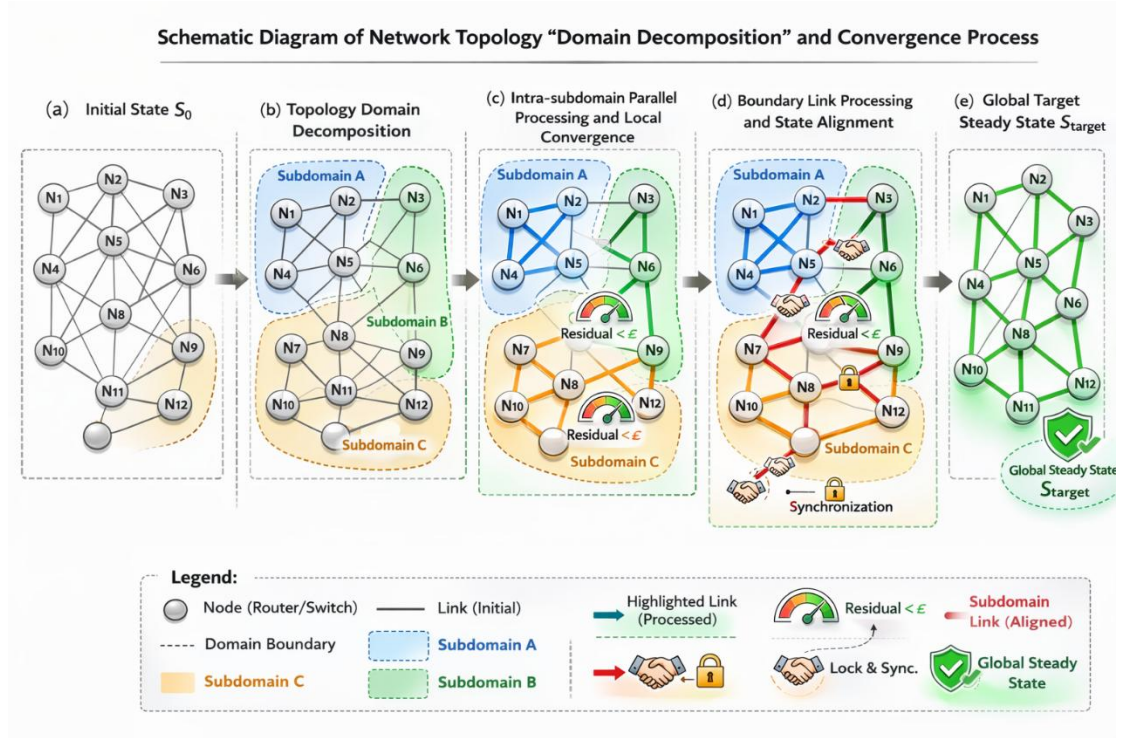


Figure 2: Schematic Diagram of Network Topology “Domain Decomposition” and Convergence Process

This figure illustrates the core “divide-and-conquer with coordinated convergence” strategy of the present invention, demonstrating a deterministic transformation mechanism from physical topology modeling to ordered logical execution.

- **(a) Initial State S_0 :**
The original network topology prior to configuration, representing a globally coupled system awaiting coordinated deployment.
- **(b) Domain Decomposition:**
This stage implements the technical means of regional decomposition. A complex, globally interdependent topology is partitioned into multiple non-overlapping subdomains. Each subdomain exhibits strong internal coupling and weak external coupling. By minimizing cross-domain dependencies, the invention transforms a monolithic global convergence problem into multiple parallelizable sub-problems, thereby establishing the structural foundation for concurrent processing.
- **(c) Intra-Subdomain Convergence:**
Within each subdomain, configuration tasks are executed independently and in parallel. Protocol states converge locally, achieving a stable intra-domain

steady state. The residual within each subdomain satisfies the convergence threshold condition ($\text{Residual} < \epsilon$), indicating local equilibrium.

- **(d) Boundary Alignment:**
After local convergence is achieved, boundary links between subdomains are processed in a coordinated and deterministic manner. This controlled boundary synchronization integrates the locally converged states into a globally consistent system state. By sequencing inter-domain coordination after intra-domain stabilization, the invention effectively mitigates control plane oscillations and instability risks commonly associated with traditional large-scale concurrent deployments.
- **(e) Global Target Steady State S_{target} :**
The entire network reaches the intended final steady state, in which all nodes and links operate in a globally synchronized and stable configuration.

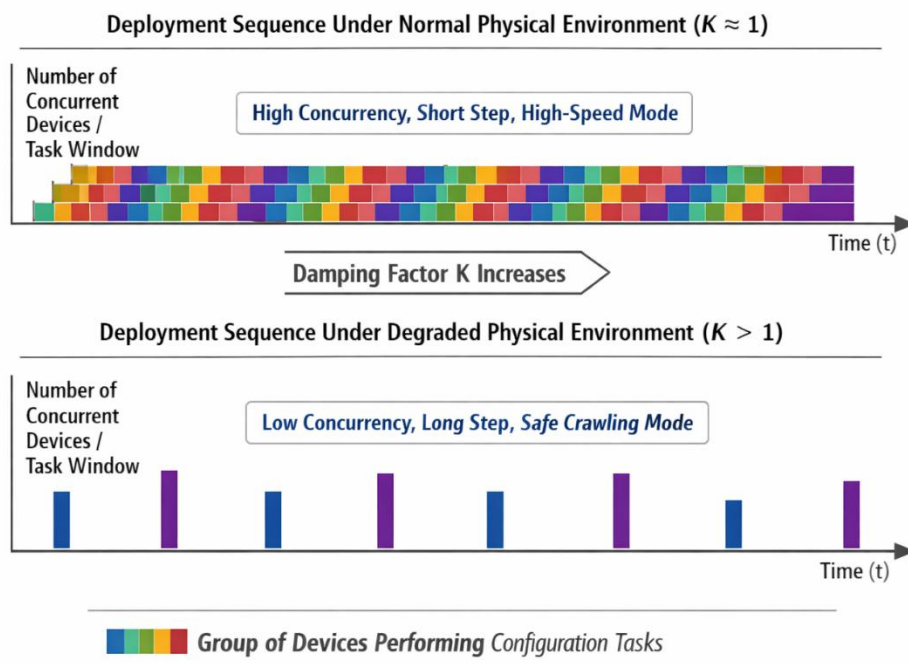


Figure 3: Comparative Diagram of Physical Field Damping Coefficient K 's Correction to Deployment Sequence

Through the upper and lower comparative graphs, the control effect of the physical-field damping coefficient is intuitively illustrated:

- **Upper Graph (Normal Mode, $K \approx 1$):**
The deployment sequence is densely scheduled with a high degree of concurrency, thereby achieving optimal execution efficiency.
- **Lower Graph (Degraded Mode, $K > 1$):**
The deployment sequence becomes more sparsely distributed, and the concurrency window is reduced. The system automatically transitions

into a “safe crawling mode,” sacrificing a portion of temporal efficiency in exchange for enhanced operational reliability.